Mittuniversitetet
MID SWEDEN UNIVERSITY

DT081A – Signal and Image Processing

Lecture 10 - Spatial domain image processing III

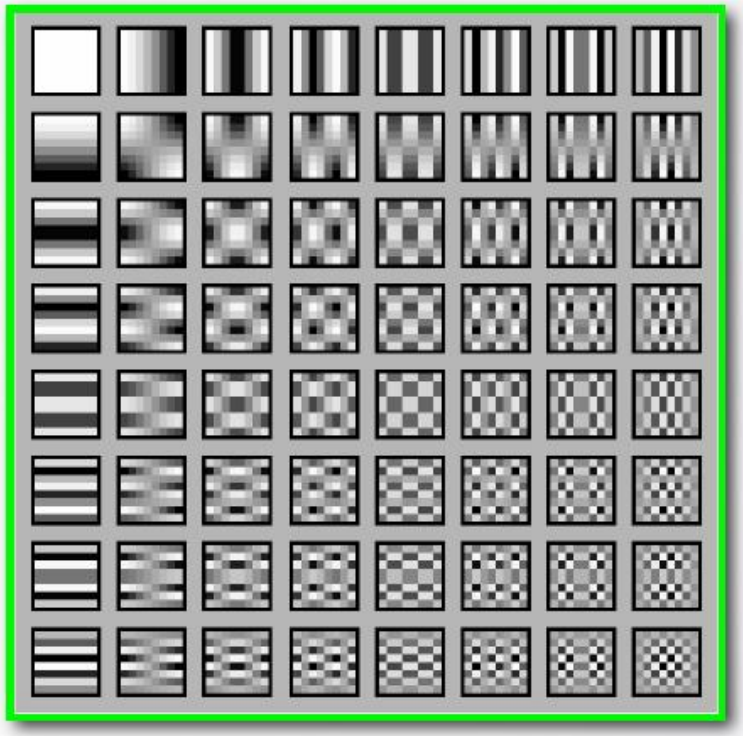


Roger Olsson (PhD)
Department of Computer and Electrical Engineering

Discrete Cosine Transform

$$F(u,v) = \frac{2\,C(u)\,C(v)}{\sqrt{MN}} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} \cos\frac{(2i+1)\cdot u\pi}{2M} \cdot \cos\frac{(2j+1)\cdot v\pi}{2N} f(i,j)$$

672	175	9	13	5	0	-1	-3
-281	-176	18	-5	-4	1	-1	1
150	28	-55	-5	-9	0	-2	0
-51	39	46	0	8	1	0	-5
1	-35	-15	15	1	0	0	-3
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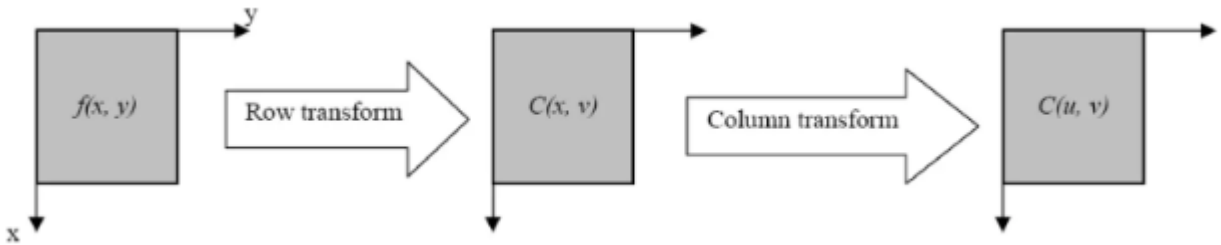


DCT – a separable base (*)

- The 2D 8x8 DCT can be separated into a sequence of two, 1D DCT steps:

$$G(i,v) = \frac{1}{2}C(v)\sum_{j=0}^{7}\cos\frac{(2j+1)v\pi}{16}f(i,j)$$

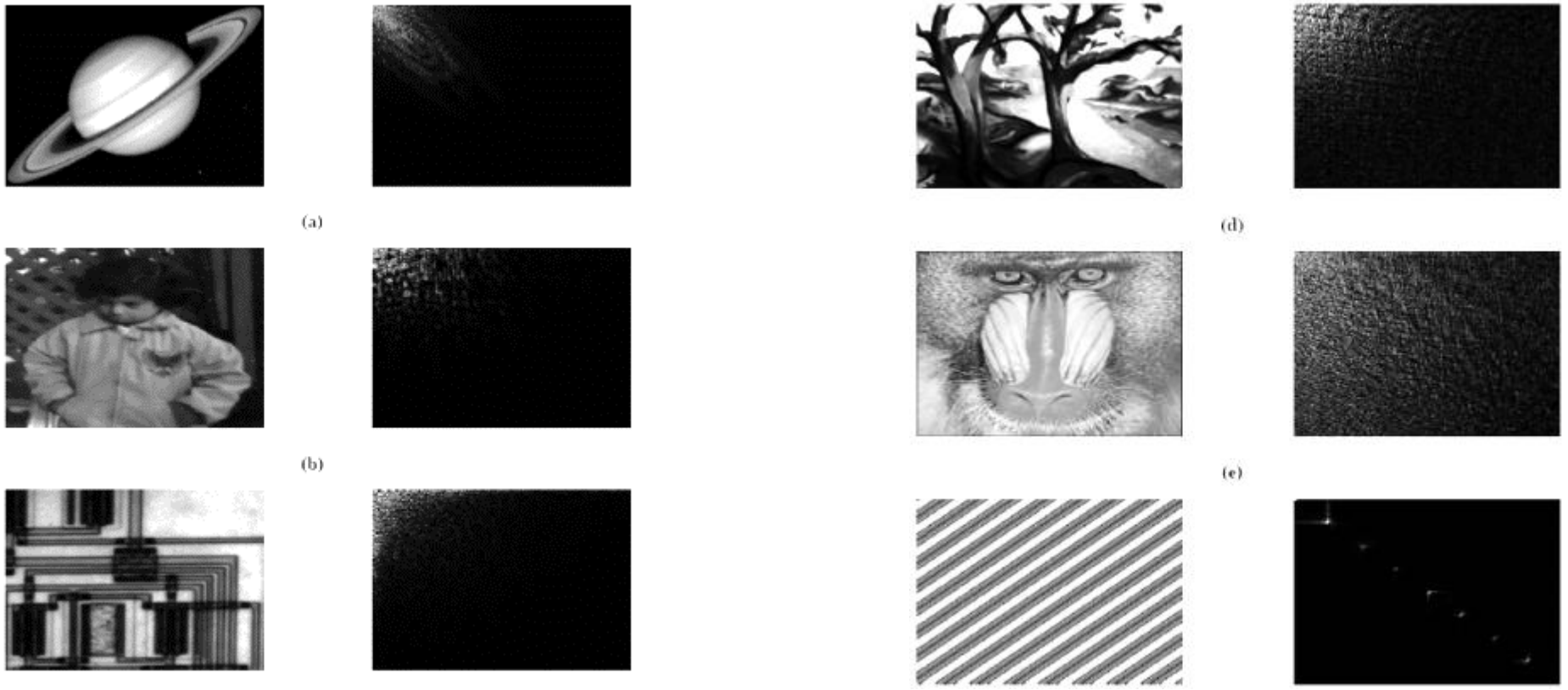
$$F(u,v) = \frac{1}{2}C(u)\sum_{i=0}^{7}\cos\frac{(2i+1)u\pi}{16}G(i,v)$$



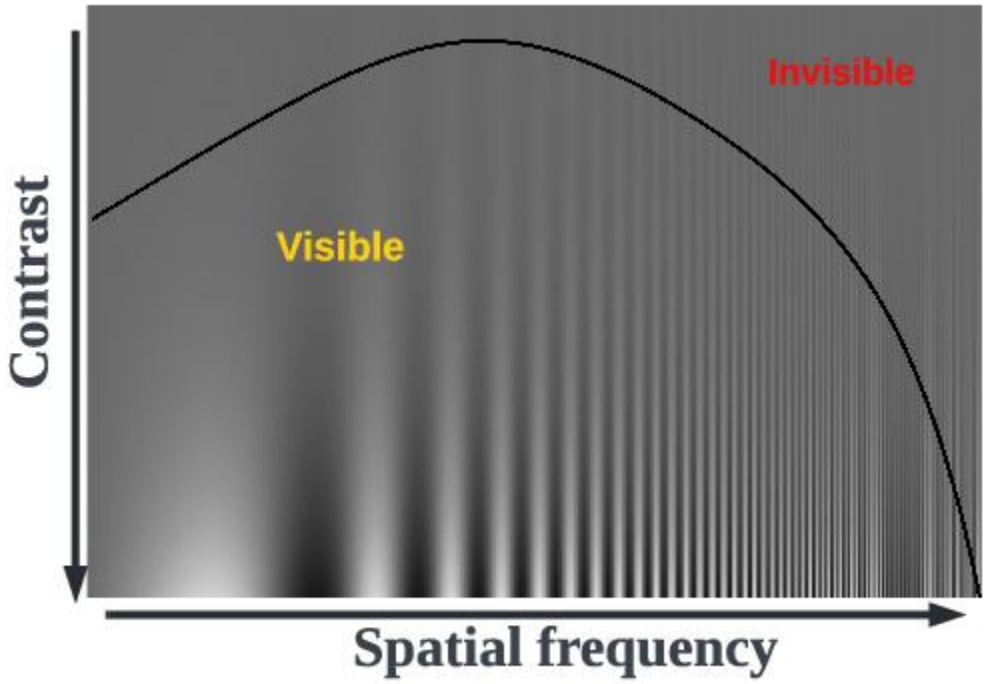
- A separable transformation can be performed much faster, less computations are required

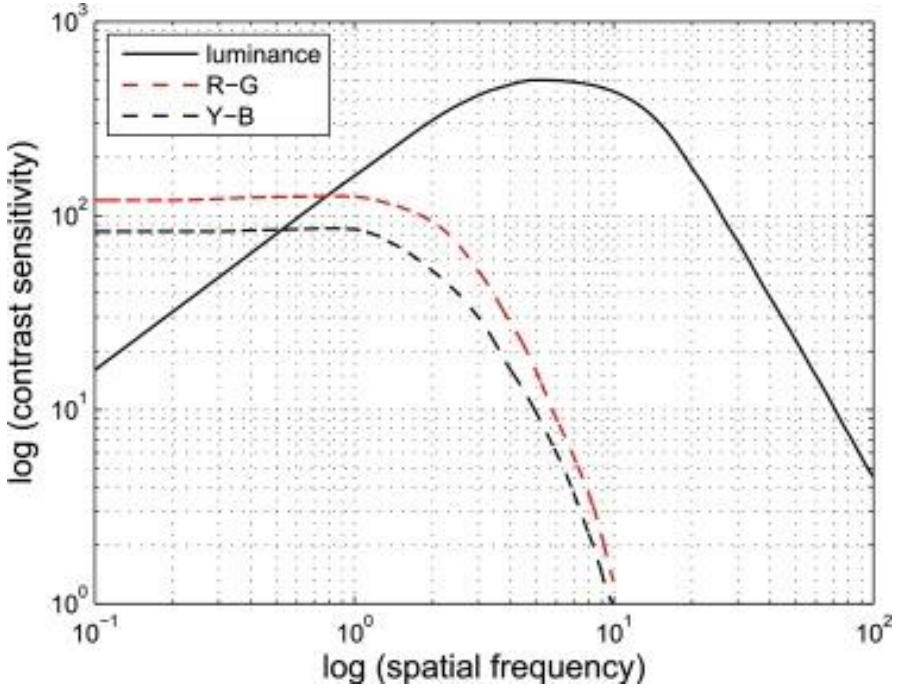
(*) as is the DFT

Why DCT? Favourable energy compaction properties



Spatial response and its low pass characteristics

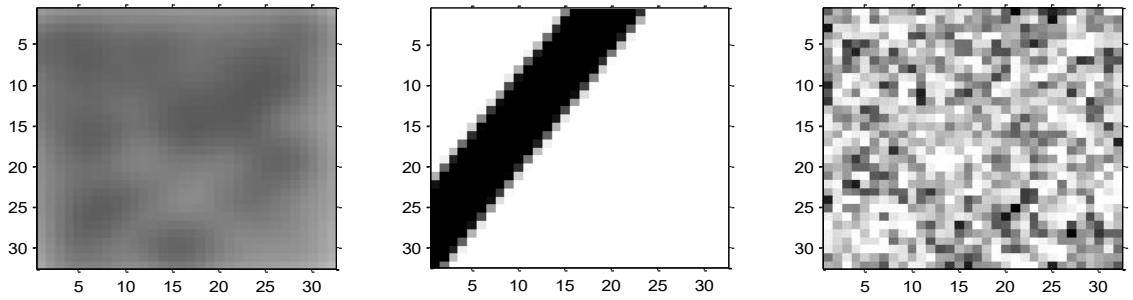




Prioritize visible details

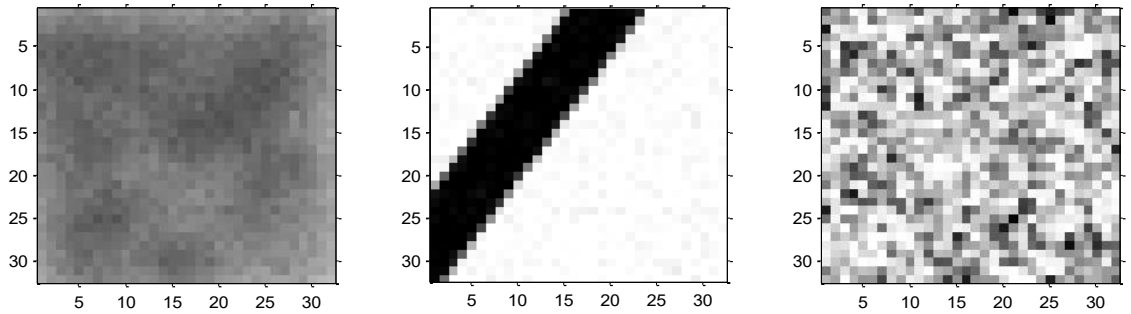
- Saturated HVS, details mask noise

No noise



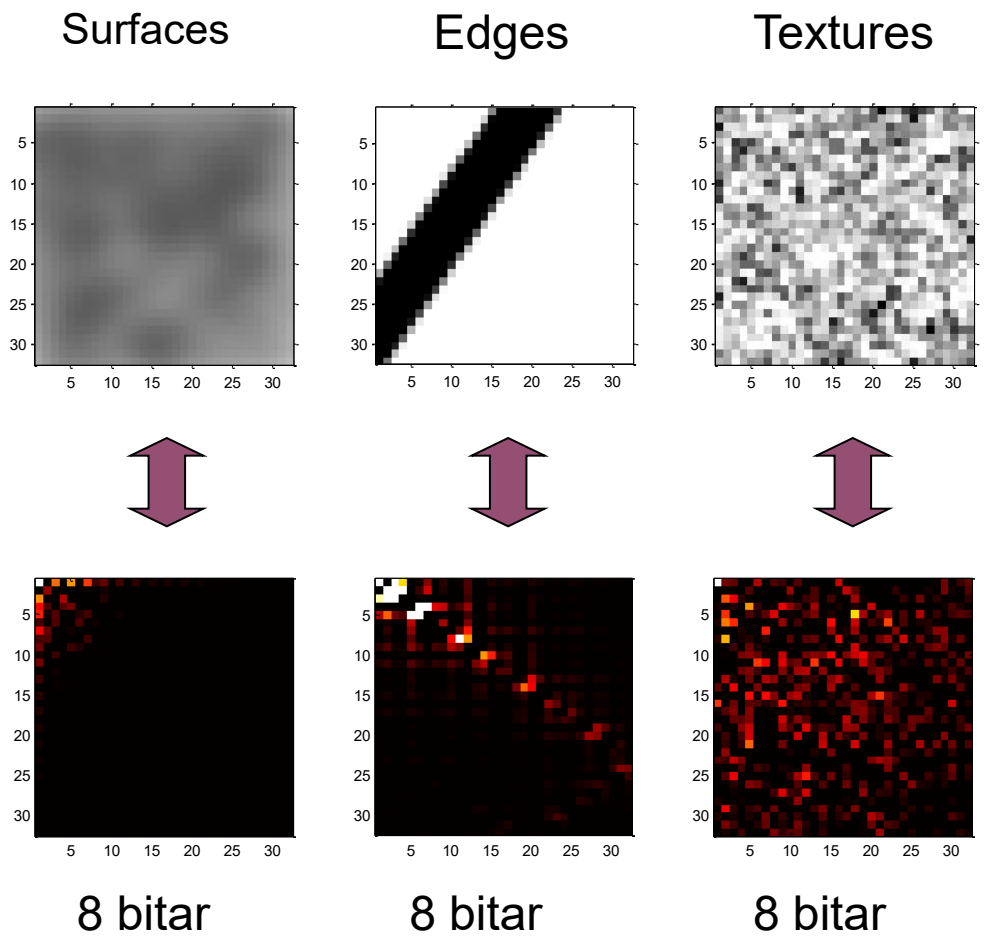
Complexity

Added noise

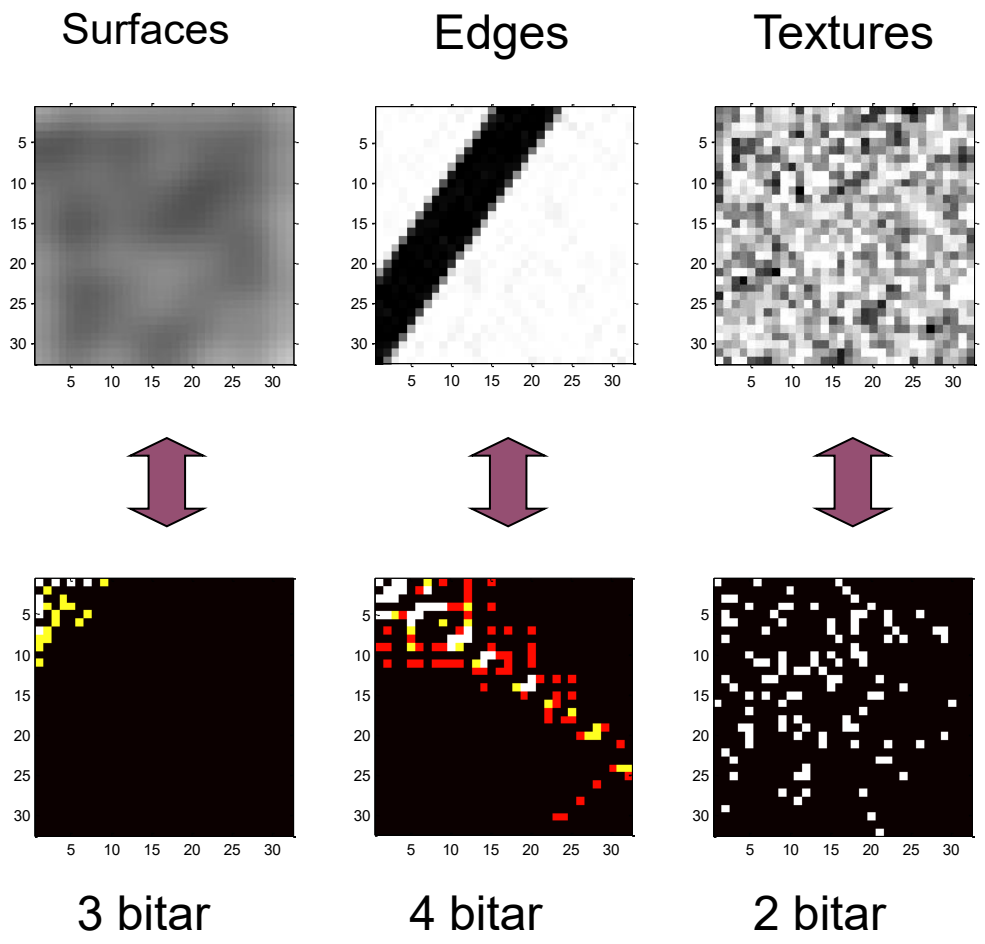


Same amount of noise in all three images

Prioritize visible details

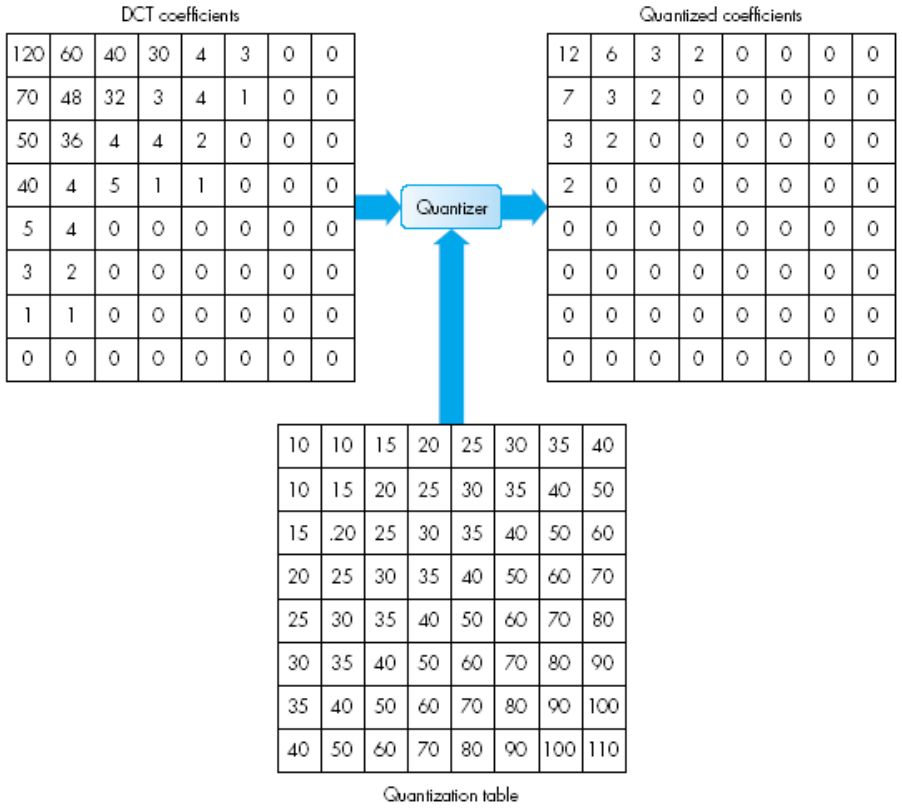


Prioritize visible details

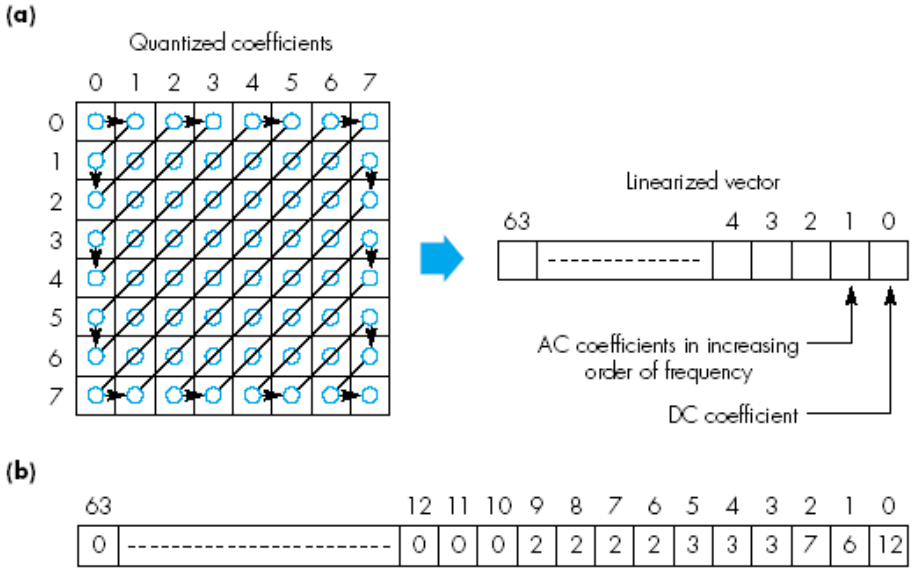


Block-wise quantization of DCT coefficients

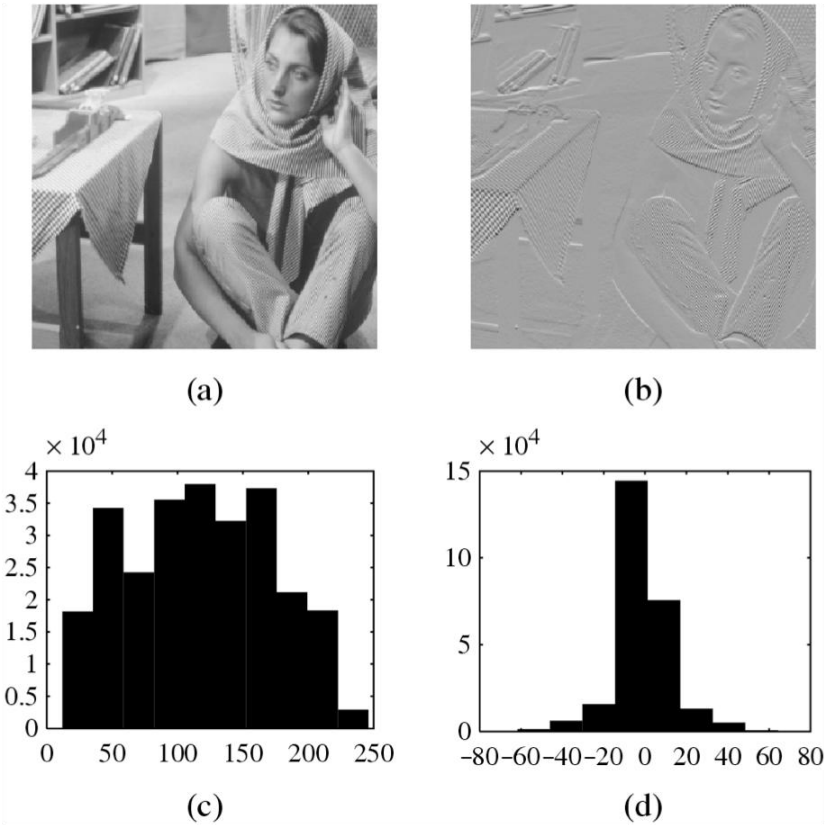
- Scaled quantization



- Vectorization for later entropy encoding



Lossless Image compression

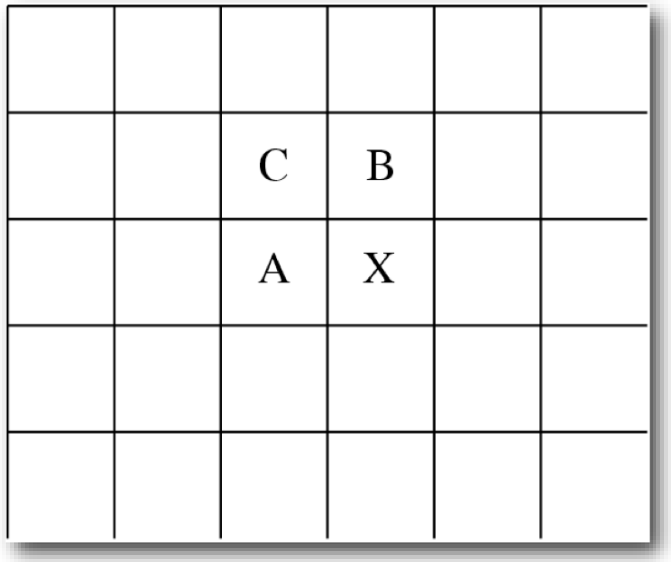


Distributions for Original versus Derivative Images. (a,b): Original gray-level image and its partial derivative image; (c,d): Histograms for original and derivative images.

Lossless JPEG compression

The Predictive method

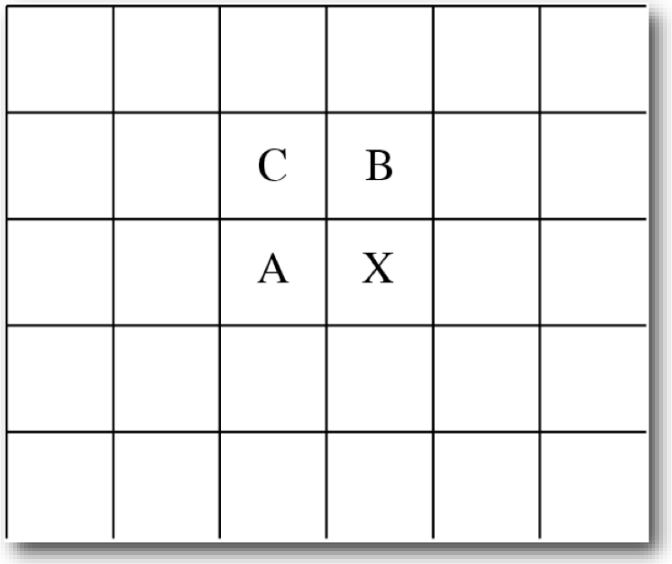
1. Forming a differential prediction: A predictor combines the values of up to three neighboring pixels (A, B, and C) as the predicted value for the current pixel, indicated by ‘X’.
2. Encoding: The encoder compares the prediction with the actual pixel value at the position ‘X’ and encodes the difference using one of the lossless compression techniques we have discussed, e.g., the Huffman coding scheme



Lossless JPEG compression

Predictors

Predictor	Prediction
P1	A
P2	B
P3	C
P4	A + B – C
P5	A + (B – C) / 2
P6	B + (A – C) / 2
P7	(A + B) / 2



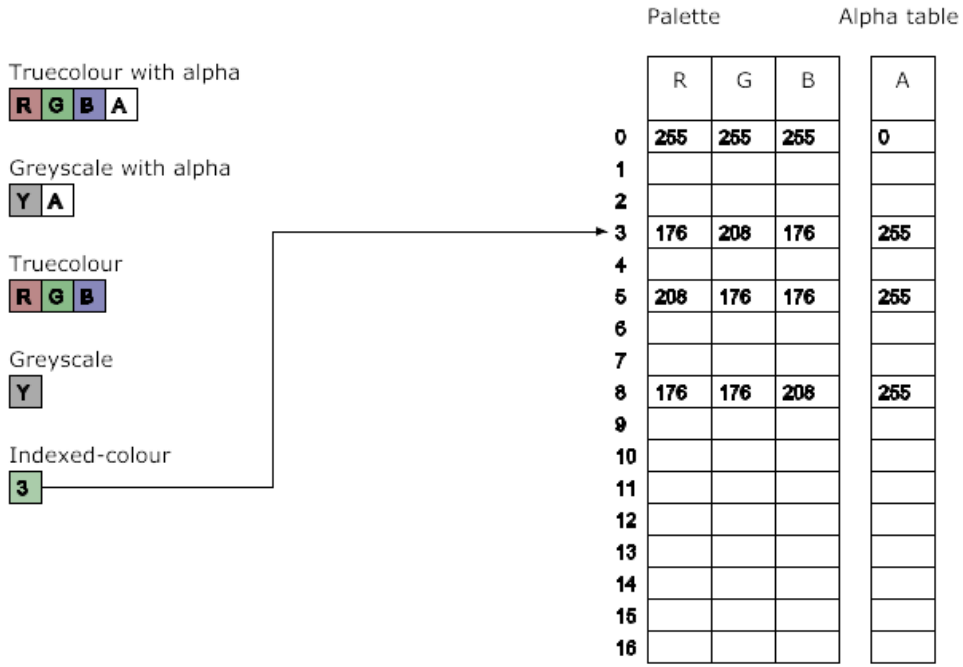
Lossless JPEG compression

Comparison with other lossless compression methods

Method	Compression Ratio			
	Lena	Football	F-18	Flowers
Lossless JPEG	1.45	1.54	2.29	1.26
Optimal Lossless JPEG	1.49	1.67	2.71	1.33
Compress (LZW)	0.86	1.24	2.21	0.87
Gzip (LZ77)	1.08	1.36	3.10	1.05
Gzip -9 (optimal LZ77)	1.08	1.36	3.13	1.05
Pack(Huffman coding)	1.02	1.12	1.19	1.00

PNG - Portable Network Graphics

- PNG was created in response to the patented LZW algorithm used in GIF, with a focus on creating a free alternative and extend functionality
- Supports palette-based (palettes of 24-bit RGB colors), greyscale, RGB, or RGBA images with up to 48 bits of color information.
- Images may contain gamma-correction information for correct display of color and alpha-channel information for control of transparency.
- Allows for 5 types of image representation



PNG encoding

Pass extraction

- image pixels can be rearranged to form several smaller images called reduced images or passes, or
- be extracted column by column, row by row.

Scanline serialization

- Each image is serialized a scanline at a time, with pixels from left to right.

Filtering

- each scanline is transformed into a filtered scanline using one of the defined filter types to prepare for compression.

Compression

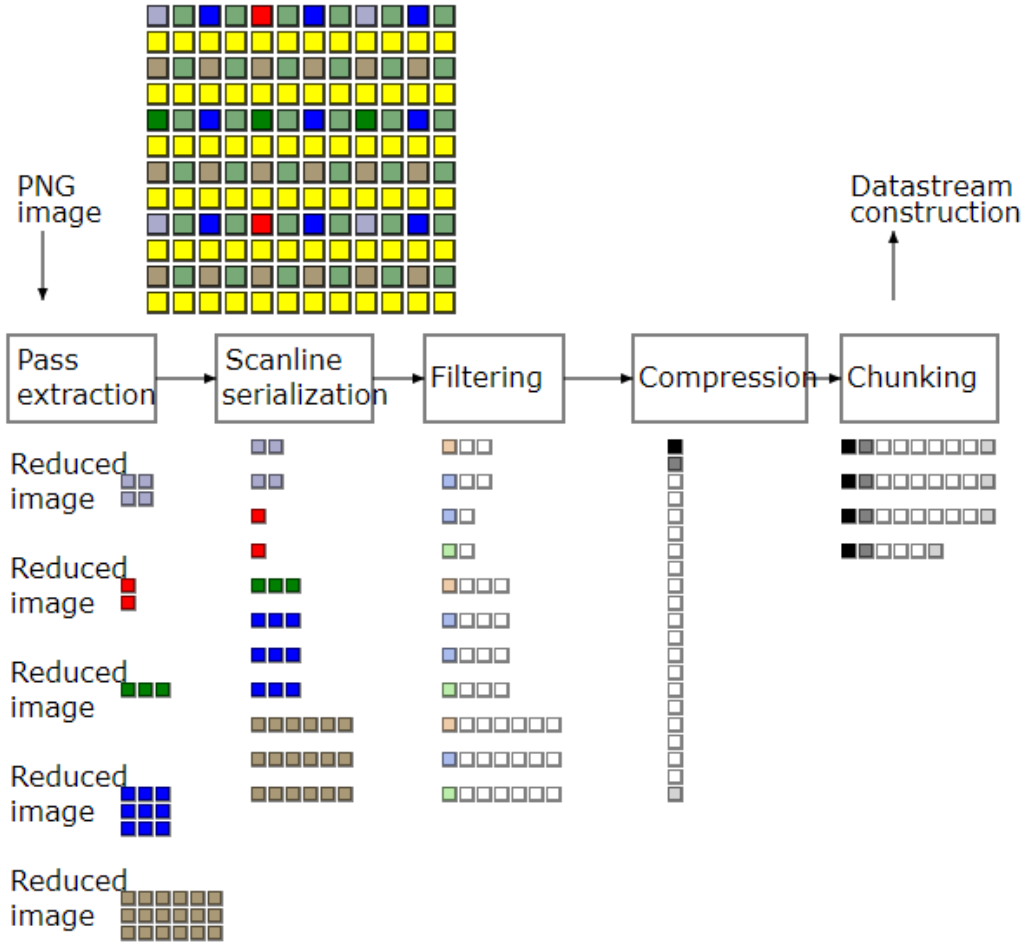
- occurs on all the filtered scanlines in the image.

Chunking

- the compressed symbols are is divided into conveniently sized chunks. An error detection code is added to each chunk.

Datastream construction

- the chunks are inserted into the datastream.



Elementary image

An image can be expressed as the linear superposition of some elementary images

Elementary image = the outer product of two vectors

Consider two $Nx1$ vectors $\mathbf{u}$ and $\mathbf{v}$

$$\boldsymbol{u}_i = \begin{pmatrix} u_{i1} \\ u_{i2} \\ \vdots \\ u_{iN} \end{pmatrix}, \boldsymbol{v}_j = \begin{pmatrix} v_{j1} \\ v_{j2} \\ \vdots \\ v_{jN} \end{pmatrix}$$

The outer product is defined as

$$\boldsymbol{u}_i \boldsymbol{v}_j^{\mathrm{T}} = \begin{pmatrix} u_{i1} \\ u_{i2} \\ \vdots \\ u_{iN} \end{pmatrix} \begin{pmatrix} v_{j1} & v_{j2} & \cdots & v_{jN} \end{pmatrix}$$

$$= \begin{pmatrix} u_{i1}v_{j1} & u_{i1}v_{j2} & \cdots & u_{i1}v_{jN} \\ u_{i2}v_{j1} & u_{i2}v_{j2} & \cdots & u_{i2}v_{jN} \\ \vdots & \vdots & \ddots & \vdots \\ u_{iN}v_{j1} & u_{iN}v_{j2} & \cdots & u_{iN}v_{jN} \end{pmatrix}$$

Which can be seen as an image

Elementary image expansion

We know that a separable linear transformation of image f can be written a

$$g = h_c^T f h_r$$

with h_c and h_r as two transforming matrices

Let's solve for f

$$(h_c^T)^{-1} g (h_r)^{-1} = (h_c^T)^{-1} h_c^T f h_r (h_r)^{-1} = f$$

$$f = (h_c^T)^{-1} g (h_r)^{-1}$$

Suppose we partition $(h_c^T)^{-1}$ and $(h_r)^{-1}$ in their respective column and row vectors, then

$$f = (\mathbf{u}_1 \quad \mathbf{u}_1 \quad \dots \quad \mathbf{u}_N) g \begin{pmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_N^T \end{pmatrix}$$

Lets now split up g into a sum of N^2 NxN matrices

$$g = \begin{pmatrix} g_{11} & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix} + \begin{pmatrix} 0 & g_{12} & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix} + \cdots + \begin{pmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & g_{NN} \end{pmatrix}$$

Elementary image expansion

Suppose we partition $(h_c^T)^{-1}$ and $(h_r)^{-1}$ in their respective column and row vectors, then

$$f = (\mathbf{u}_1 \quad \mathbf{u}_1 \quad \dots \quad \mathbf{u}_N) g \begin{pmatrix} \mathbf{v}_1^{\mathrm{T}} \\ \mathbf{v}_2^{\mathrm{T}} \\ \vdots \\ \mathbf{v}_{\mathrm{N}}^{\mathrm{T}} \end{pmatrix}$$

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Then *f* can be written as

$$f = \sum_{i=1}^{N} \sum_{j=1}^{N} \mathrm{g_{ij}} \boldsymbol{u}_i \boldsymbol{v}_{\mathrm{j}}^{\mathrm{T}}$$

Elementary image conclusion

Any image f may be expressed by a weighted sum of elementary images defined using the outer product

$$f = \sum_{i=1}^{N} \sum_{j=1}^{N} \mathrm{g_{ij}} \boldsymbol{u}_i \boldsymbol{v}_\mathrm{j}^\mathrm{T}$$

The choice of vectors in $\boldsymbol{u}_i \boldsymbol{v}_\mathrm{j}^\mathrm{T}$ determines the specific type of image representaion

- Fourier Transform (using complex exponential basis functions)
- Principal Component Analysis (using eigenvectors of the covariance matrix of f)
- Wavelet Transform (using wavelet basis functions)
- Singular Value Decomposition (using left and right singular vectors of f)
- …

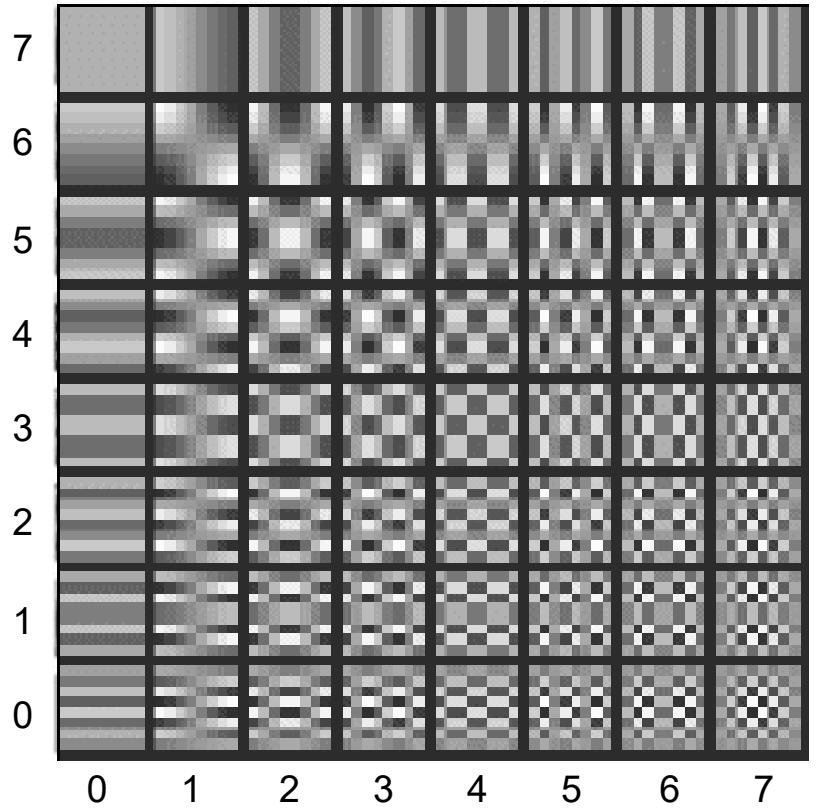
Fourier transform

The inverse 2D discrete Fourier transform is

$$\hat{g}(q,p) = \frac{1}{\sqrt{MN}} \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} \hat{g}(m,n) e^{2\pi j\left(\frac{km}{M}+\frac{ln}{N}\right)}$$

The fourier transform of an MxN image g is

$$\hat{g}(m,n) = \frac{1}{\sqrt{MN}} \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} g(k,l) e^{-2\pi j\left(\frac{km}{M}+\frac{ln}{N}\right)}$$



Fourier transform

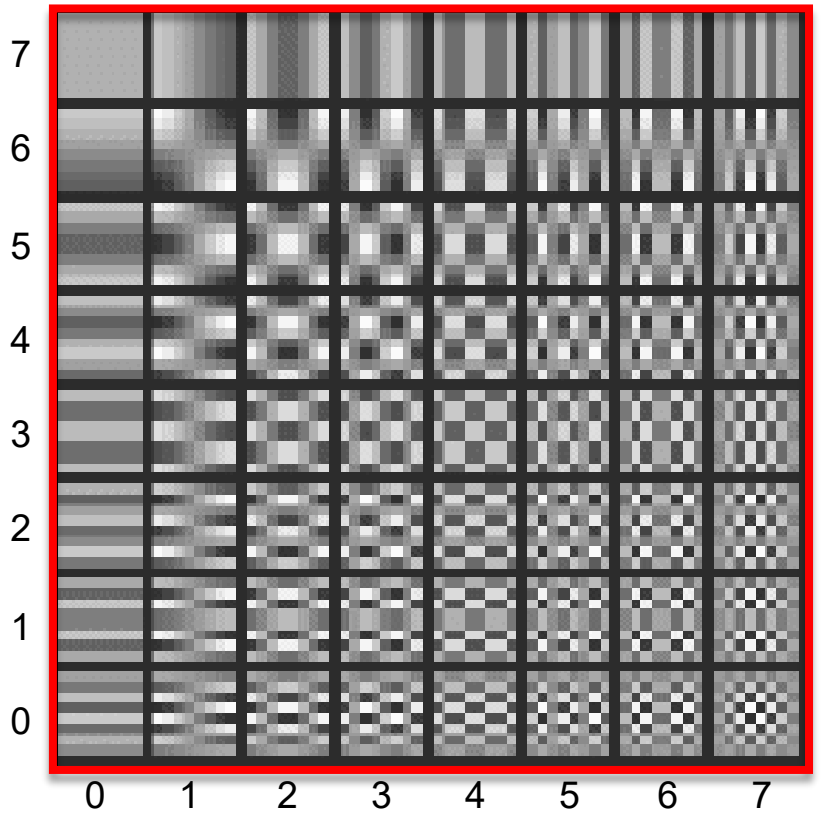
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Example for an 8x8 image



Fourier transform in matrix format

- Construct a unitary (*) matrix U with elements

$$U_{x\alpha} = \frac{1}{\sqrt{N}} e^{-j\frac{2\pi x\alpha}{N}}$$

Where x takes the values $0, 1, , N-1$ along each column and α takes the same calues along each row. Note that U then becomes symmetric

$$U^T = U$$

The 2D discrete Fourier transform of image g then becomes

$$\hat{g} = UgU$$

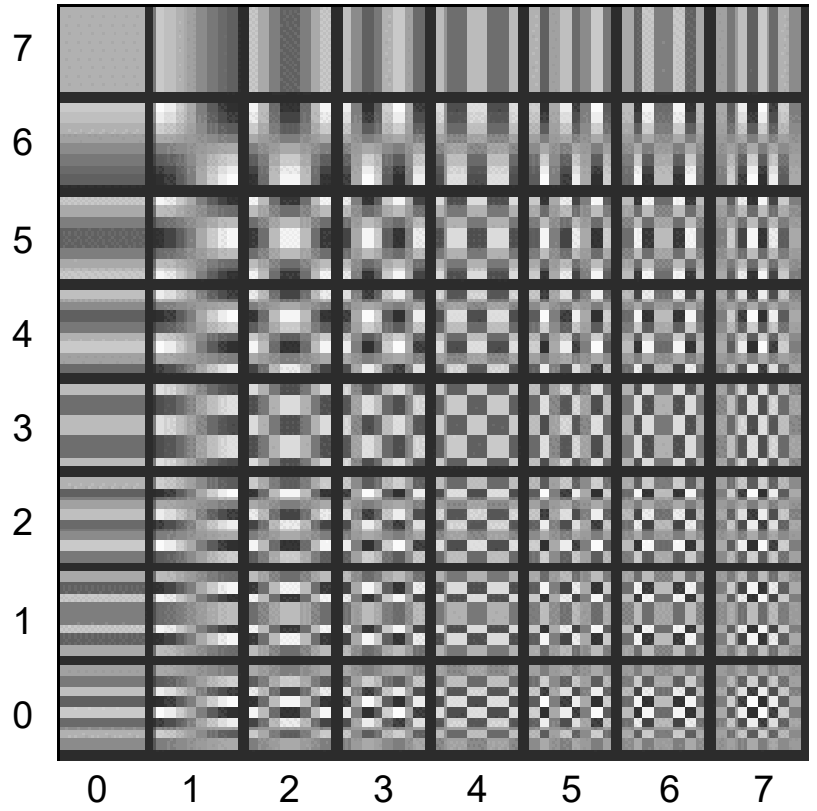
(*) a matrix is called unitary if its inverste is the complex conjugate of its transpose

$$UU^{\mathrm{T}*} = I$$

DCT

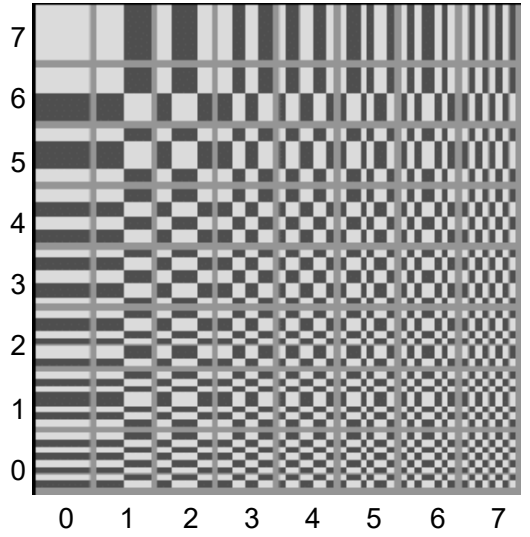
- The DCT is widely used in image compression (e.g., JPEG).
- It transforms image data into a sum of cosine basis functions of varying frequencies.
- The DCT focuses on energy compaction, meaning most of the image's information is concentrated in a few low-frequency components.
- Unlike DFT, the DCT works with real numbers, making it computationally efficient.
- The unitary matrix U is defined using

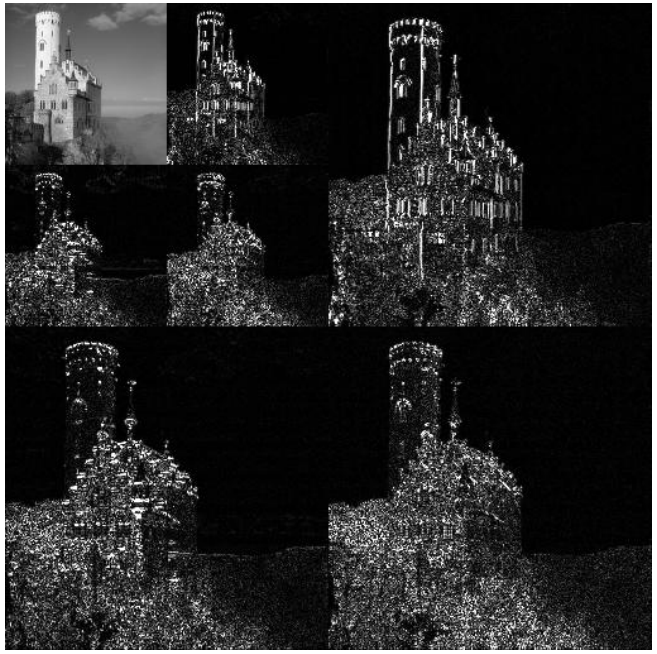
$$U_{x\alpha} = \sqrt{\frac{2}{N}} cos\left(\frac{\pi(2x+1)\alpha}{2N}\right)$$



Wavelet

- Haar wavelet is a simple and widely used wavelet for image compression and analysis.
- It decomposes an image into different scales and resolutions by applying both high and low pass filters.
- Haar wavelet allows for the representation of image details at multiple levels of resolution, facilitating analysis.
- The Haar wavelet works by capturing both large-scale structures and fine-grained details.

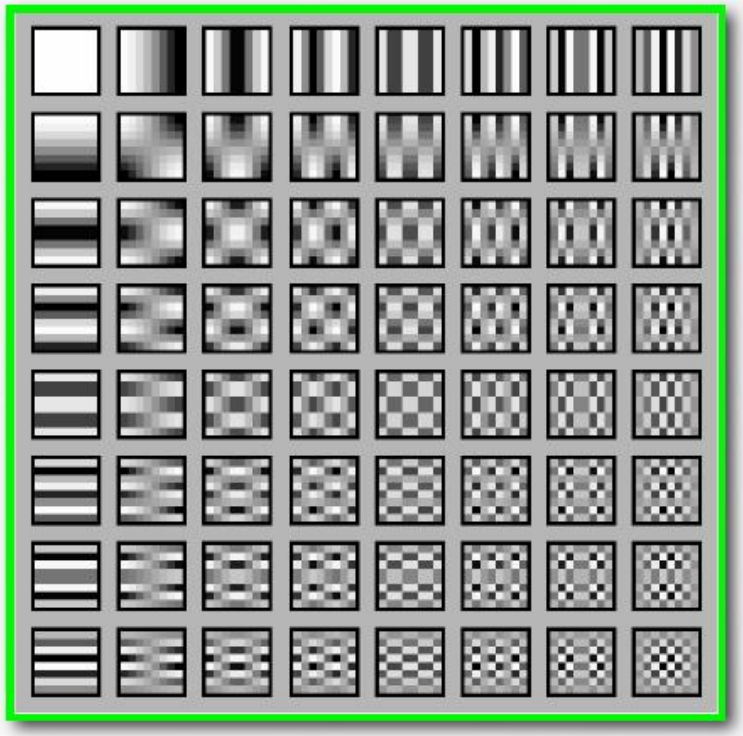




Discrete Cosine Transform

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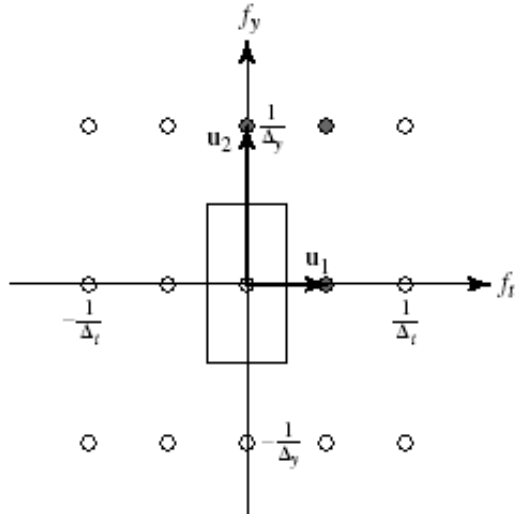


1D Nyquist Sampling Theorem revisited

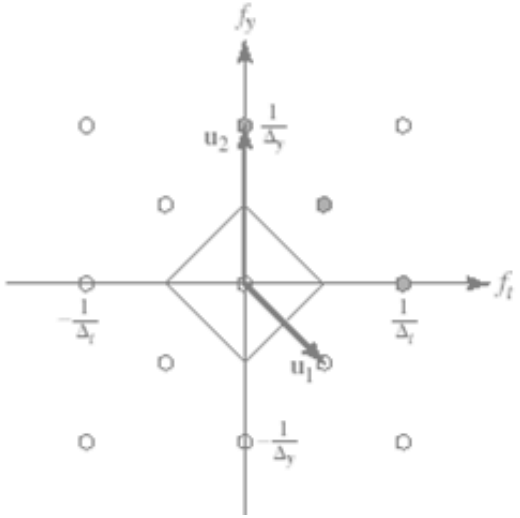
- Given a band-limited signal with maximum frequency f_{max}, it can be sampled with a sampling rate $f_s >= 2 f_{max}$. The original continuous signal can be reconstructed (interpolated) from the samples exactly, by using an ideal low pass filter with cut-off frequency at $f_s /2$.
- Practical interpolation filters: replication (sample-and-hold, 0th order), linear interpolation (1st order), cubic-spline (2nd order)
- Given the maximally feasible sampling rate f_s, the original signal should be bandlimited to $f_s /2$, to avoid aliasing. The desired prefilter is an ideal low-pass filter with cut-off frequency at $f_s /2$.
- Prefilter design: Trade-off between aliasing and loss of high frequency

Extension to Multi-dimensions

- If the sampling grid is aligned in each dimension (rectangular in 2-D) and one performs sampling in each dimension separately, the extension is straightforward:
  - Requirement: $f_{s,i} <= f_{max,i}/2$
  - Interpolation/pre-filter: ideal low-pass in each dimension



- If the sampling grid is an arbitrary lattice, the support region of the signal spectrum becomes more complex for this course



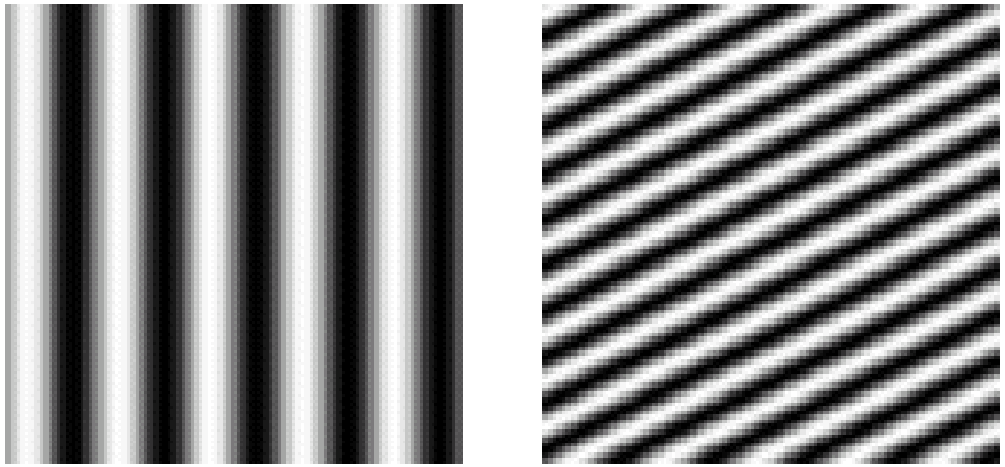
Spatial Frequency

- Spatial frequency measures how fast the image intensity changes in the image plane
- Spatial frequency can be completely characterized by the variation frequencies in two orthogonal directions (e.g horizontal and vertical)
  - f_x: cycles/horizontal unit distance
  - f_y : cycles/vertical unit distance
- It can also be specified by magnitude and angle of change

$$f_m = \sqrt{f_x^2 + f_y^2}$$

$$\theta = \arctan\left(f_y / f_x\right)$$

Illustration of Spatial Frequency

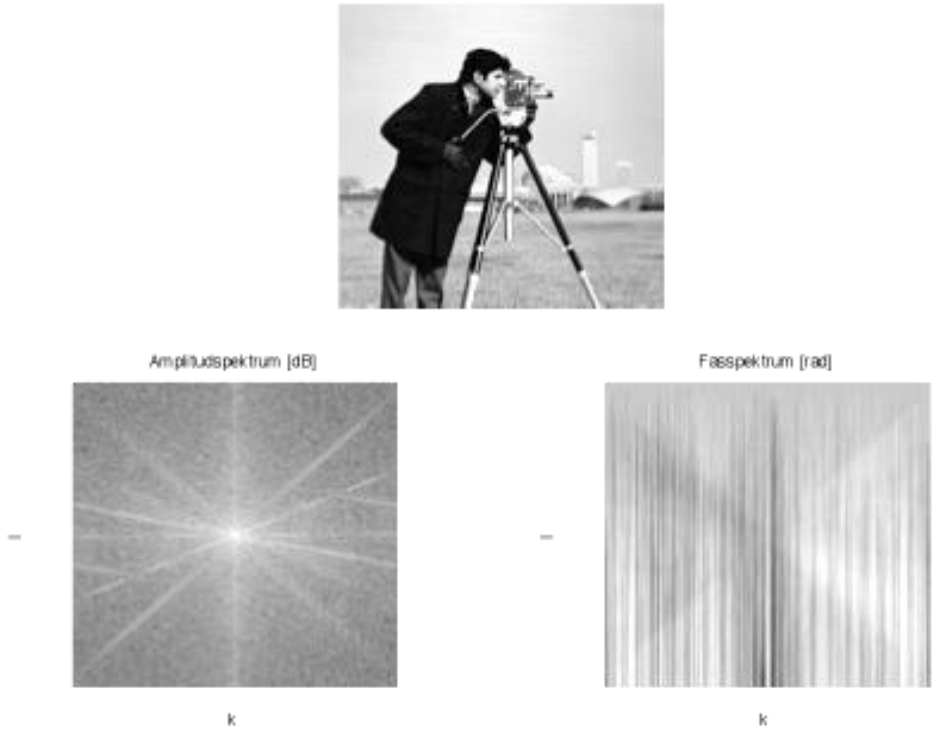


(a) (b)

Figure 2.1 Two-dimensional sinusoidal signals: (a) $(f_x, f_y) = (5, 0)$; (b) $(f_x, f_y) = (5, 10)$. The horizontal and vertical units are the width and height of the image, respectively. Therefore, $f_x = 5$ means that there are five cycles along each row.

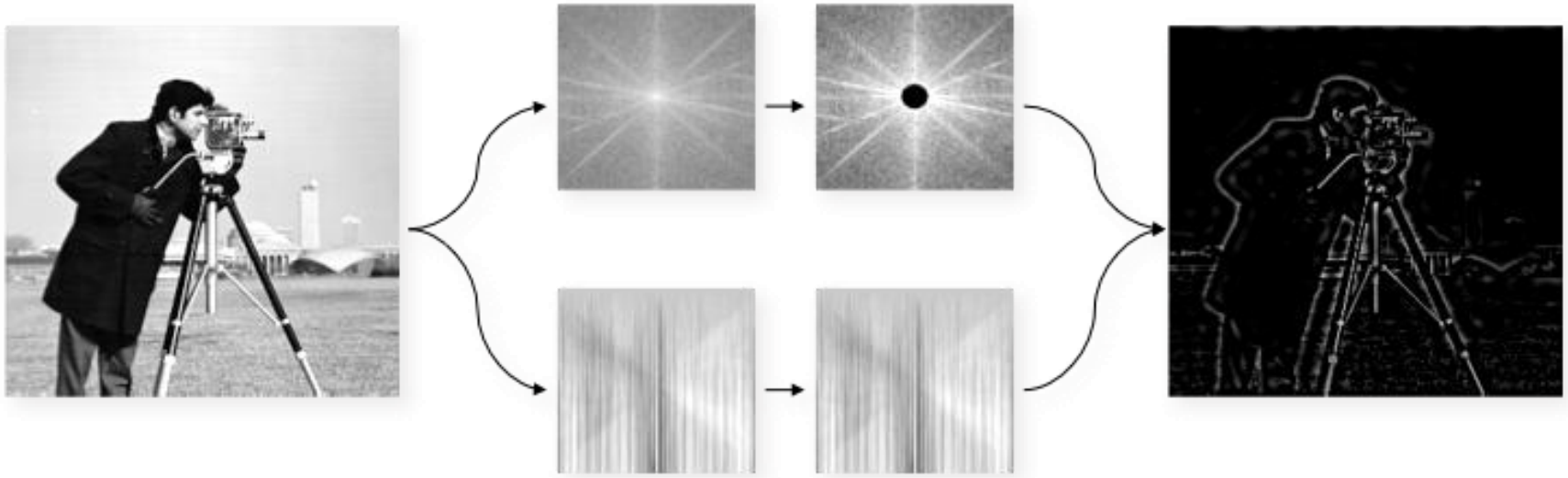
Spatial Frequency

- Spatial frequency = the rate of pixel value change over a specific length (# of pixels, mm, inches etc.)
- The frequency representation of an image consists of
  - Amplitude spectrum
  - Phase spectrum



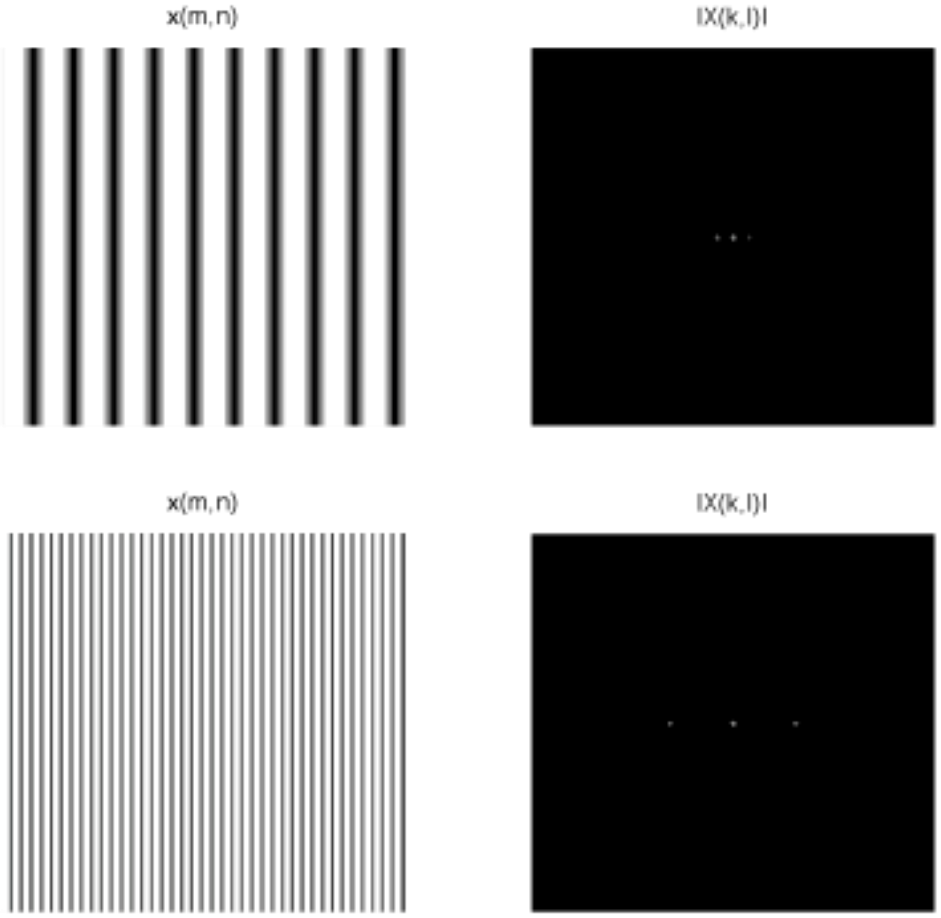
Spatial frequency domain

- Processing can be conducted in the spatial frequency domain as well as in the pixel domain of the original signal



Spatial Frequency Domain

- Same direction, different frequencies



Spatial Frequency Domain

- Same direction, same frequency but different edge sharpness

